



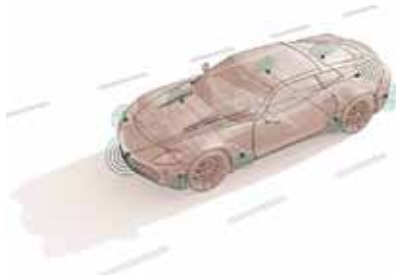
pixel CDS (correlated double sampling) and advanced dual light guides to achieve good SNR at low light, with low crosstalk for crisp images. The sensors employ a low-noise, global-shutter pixel technology to offer powerful solutions and enable compact mobile designs for many applications such as smart wearable, IoT, Robotics, etc.

Teledyne

<https://imaging.teledyne-e2v.com>

Long-range antenna for ADAS

The HUBER + SUHNER 3D wave-guide antenna is the first long-range 3D metallised plastic radar antenna designed for advanced driver-assistance systems (ADAS). The antenna offers ultra-low losses for the object



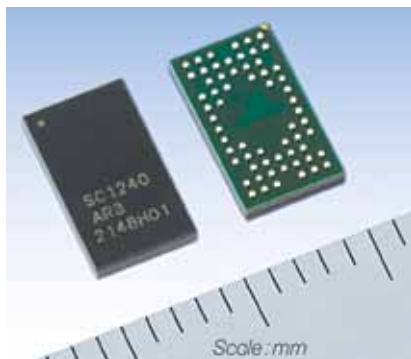
detection whilst meeting automotive standards. It offers improved efficiency, pattern stability, and impedance bandwidth at competitive manufacturing cost.

HUBER SUHNER

<https://www.hubersuhner.com/en>

Range and angle-calculating sensors

The SC1240 series radio-wave ranging sensors uses the 60GHz band with a built-in signal processing circuit for



detecting the position and movement of a person with maximum accuracy. It is the industry's first sensor with a built-in range-finding and an angle-calculating signal processing circuit. It uses 3D position information without the need for high frequency or advanced signal processing technology. It includes multiple antennas, wireless circuitry, A/D converters, FIFO memory, SPI interface, a high-performance sequencer that flexibly changes the duty cycle and controls power consumption, and a signal processing circuit for high-performance distance measurement and angle calculation.

Socionext

<https://socionextus.com>

Touch-sensing chip

The AX54A-Force chip adds touch functionality to surfaces, touch button clusters, and other solutions. It features



an integrated force-sensing technology which provides accurate and highly sensitive force-sensing data to the user interface (UI). The AX54A offers a high signal-to-noise ratio (SNR), enabling precise detection with high precision. It can accurately take inputs from 0.1 Newton (10g) to 20 Newton (2000g). It utilises a unique integrated force-sensing technology, which enables the detection of a wide range of forces, thus making it suitable for a wide range of applications.

Touchnetix

<https://www.touchnetix.com>

TOLL-packaged SiC MOSFET

Onsemi NTBL045N065SC1 is the world's first TOLL-packaged 650V SiC MOSFET. It has a VDSS rating of 650V with a typical RDS(on) of just 33mΩ and a maximum drain current (ID)



of 73A. The transistor can address the rapidly growing need for high-performance switching devices that are suitable for designs with high levels of power density. The module comes in a compact package with a footprint of just 9.90 mm x 11.68 mm and offers 30% savings in the PCB area over a D2PAK package. At a profile of just 2.3mm, it occupies 60% less volume than a D2PAK package.

Onsemi

<https://www.onsemi.com>

DC-DC controller for LED lights

Infineon's Litix TLD6098-2ES is a dual-channel, DC-DC controller which can drive a full LED headlamp without the requirement of a microcontroller. It can operate the four standard LED front light functions, which include high beam, low beam, daytime running



lights, and turn signal. It can also be used as a voltage supply for controlling the animation in exterior LED lighting. The controller can work with a wide input voltage range of 4.5V to 60V. Each channel of TLD6098-2ES offers both analogue and PWM dimming for precise LED brightness control.

Infineon Technologies

<https://www.infineon.com>